

Mutual Fund Vs Direct Equity Investment Trend Analysis in India Using AMFI and Market Data

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ABSTRACT

The investment market in India offers several options for investors, among which mutual funds and direct equity are two important choices. Mutual funds provide professionally managed investment opportunities, while direct equity allows investors to invest directly in listed companies. This study analyzes the investment trends of mutual funds and direct equity in India using available financial data. The mutual fund data is collected from the Association of Mutual Funds in India (AMFI), while the direct equity data covers activities related to major Indian stock exchanges. Python, Pandas, NumPy, SQL, Matplotlib, Seaborn, and Power BI are used for data storage, processing, analysis, visualization, and dashboard development. The study focuses on important parameters such as Assets Under Management (AUM), net inflow, number of folios, equity settlement value, and exchange-wise equity value. The main objective is to identify yearly and monthly investment trends and provide a comparative analysis of mutual fund and direct equity activity in India.

Keywords: Mutual Funds, Direct Equity, AMFI, AUM, Net Inflow, Investment Trends, Financial Data Analysis, Python, SQL, Power BI.

I. INTRODUCTION

Investment plays an important role in personal financial planning and economic development. In India, investors have access to different financial instruments such as bank deposits, mutual funds, stocks, bonds, and other market-linked products. Among these, mutual funds and direct equity are widely used investment options.

A mutual fund collects money from multiple investors and invests it in a portfolio of securities according to the investment objective of the scheme. AMFI provides industry data and scheme-level information including AUM and folio-related statistics. Direct equity investment refers to purchasing shares of individual companies through the stock market. The Indian equity market includes major exchanges such as the National Stock Exchange (NSE) and Bombay Stock Exchange (BSE).

This study analyzes available financial data for the project period and compares mutual fund and direct equity activity using data analytics techniques. The purpose is descriptive trend analysis rather than investment advice.

II. OBJECTIVES OF THE STUDY

1. To analyze the yearly trend of mutual fund Assets Under Management (AUM).
2. To analyze mutual fund net inflow trends.
3. To study the trend in mutual fund folios.
4. To analyze direct equity settlement value.
5. To compare BSE and NSE equity values.
6. To compare mutual fund and direct equity trends using a normalized index.
7. To use Python and SQL for financial data analysis.

8. To develop an interactive Power BI dashboard for presenting the results.

III. DATA SOURCES

The study uses secondary financial data collected from financial market sources. Mutual fund data is obtained from AMFI, which publishes monthly and other industry datasets covering financial metrics such as AUM, folio counts, inflows, and related scheme information. Direct equity data used in the project contains monthly observations for BSE and NSE equity values and total equity settlement value. The analysis uses the common period available in the project datasets.

The mutual fund dataset contains fields including Year, Month, Scheme Name, AUM, Net Inflow, Number of Folios, and Number of Schemes. The direct equity dataset contains Year, Month, BSE Equity Value, NSE Equity Value, and Total Equity Settlement Value.

IV. METHODOLOGY

The proposed methodology consists of data collection, data storage, data cleaning, exploratory data analysis, and visualization.

A. Data Collection: The required mutual fund and direct equity datasets are collected from financial market sources and stored in CSV format for processing.

B. Data Storage Using MySQL: MySQL is used to store the collected datasets in structured tables. SQL queries are used to retrieve records, perform aggregations, and support analysis.

C. Data Cleaning: Python and Pandas are used to check missing values, remove unwanted category or note rows, remove duplicate records, check data types, remove unnecessary columns, and generate statistical summaries.

D. Exploratory Data Analysis: Basic EDA is performed using Pandas and NumPy. The analysis includes year-wise average AUM, year-wise average net inflow, month-wise average AUM, year-wise direct equity settlement value, BSE versus NSE equity value, top mutual fund schemes by average AUM, and positive versus negative net inflow records.

E. Visualization: Matplotlib and Seaborn are used for charts, while Power BI is used to create an interactive dashboard containing key indicators and trends.

V. DATA ANALYSIS

A. Mutual Fund AUM Trend: Assets Under Management represents the value of assets managed by mutual fund schemes. The yearly average AUM is calculated as $\text{Average AUM}_y = (\text{sum of AUM observations for year } y) / N_y$, where N_y is the number of observations for year y . A line chart is used to observe changes across the study period.

B. Mutual Fund Net Inflow Trend: Net inflow represents the difference between money entering and leaving mutual fund schemes during a period. The study calculates yearly average net inflow and counts positive and negative net inflow records.

C. Mutual Fund Folio Trend: The number of folios is analyzed on a yearly basis to observe changes in the number of investor accounts represented in the dataset.

D. Direct Equity Trend: Direct equity activity is analyzed using total equity settlement value. Year-wise average settlement value is calculated and visualized using a line chart.

E. BSE and NSE Comparison: Average BSE and NSE equity values are compared using a bar chart. A Power BI share visualization is also used to show the relative contribution of the two exchange-value fields within the analyzed dataset.

VI. MUTUAL FUND AND DIRECT EQUITY COMPARISON

Direct comparison of AUM and equity settlement values in their original monetary units is not treated as a like-for-like comparison because the measures represent different financial quantities. Therefore, a normalized index is used to compare the direction and relative change of the two series.

$Index_y = (Value_y / Value_{2024}) \times 100$. The base year 2024 is represented as 100. This normalization allows the movement of the two indicators to be displayed on a common scale without treating their original monetary values as equivalent.

VII. TECHNOLOGIES USED

Technology	Purpose
Python	Data analysis and processing
Pandas	Data cleaning and manipulation
NumPy	Numerical operations
Matplotlib	Data visualization
Seaborn	Statistical visualization
MySQL	Data storage and SQL analysis
Power BI	Interactive dashboard
Jupyter Notebook	Exploratory data analysis

VIII. RESULTS AND DISCUSSION

The analysis provides a structured view of mutual fund and direct equity activity during the selected study period. The mutual fund analysis examines changes in AUM, net inflow, and folios, while the direct equity analysis examines equity settlement activity and BSE/NSE equity values.

The normalized comparison index provides a common scale for observing relative movement between the two indicators. The Power BI dashboard presents Total AUM, Average Net Inflow, Average Direct Equity Value, Total Records, Mutual Fund versus Direct Equity Trend Index, Mutual Fund Net Inflow Trend, BSE versus NSE Equity Value Trend, BSE versus NSE Equity Value Share, and Mutual Fund Folio Trend.

The findings are descriptive and depend on the period and variables included in the project datasets. They are intended to demonstrate financial data analytics rather than provide investment recommendations.

IX. LIMITATIONS

1. The analysis is based on the available secondary datasets.
2. Mutual fund AUM and direct equity settlement value represent different financial measures and should not be interpreted as identical quantities.
3. Demat account counts should not automatically be treated as the number of unique direct-equity investors because one investor may hold multiple accounts.
4. The study focuses mainly on trend analysis and does not evaluate individual investor returns.

5. Market activity can be affected by economic conditions, interest rates, regulations, corporate events, and other factors outside the scope of this analysis.

X. CONCLUSION

This study presents a data-driven analysis of mutual fund and direct equity investment activity in India. Mutual fund data from AMFI and direct equity market data are processed using SQL and Python-based analytical techniques. The study examines AUM, net inflow, folios, equity settlement value, and exchange-wise equity activity. Data cleaning and exploratory analysis are performed using Pandas and NumPy, while Matplotlib and Seaborn are used for visualization. Power BI provides an interactive platform for presenting the analyzed results. The normalized trend index allows the movement of the two indicators to be compared on a common base without treating their original financial values as equivalent. Overall, the project demonstrates how financial datasets can be transformed into structured analytical information using modern data analytics tools.

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